MIL-M-28787/191B(EC)
23 July 1980
SUPERSEDING
MIL-M-28787/191A(EC)
15 November 1979

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SAMPLE AND HOLD, ANALOG, KEY CODE PEE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic sample and hold module.

1.2 Classification This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirments . Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.33 failures per million operating hours when predicted in accordance with MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code The module key code shall be PEb.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A21 through A40. Contact pins A1 through A20 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A

3 5.6 Depth The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/191-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4 QUALITY ASSURANCE PROVISIONS

4 1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4 2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

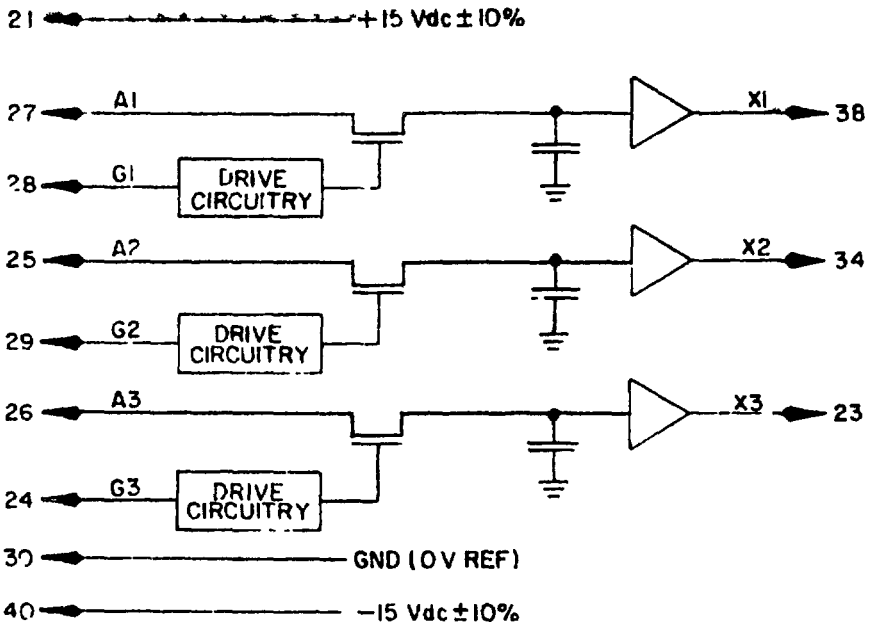
4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4 2 Order of testing. Test conditions A through S shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

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NOTES
1. A = Analog input.
   G = Gate input.
   X = Analog output.
2. Pins 1 through 20, 22, 31, 32, 33, 35, 36, 37, and 39 are unused.
3. Each unused pin shall be isolated from all other pins.

FIGURE 1. PEE module function diagram, sample and hold.

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★ TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation	II	S	10	∞	1 0	∞	10	∞	1.0	∞	MΩ
B1-B3	Output short circuit duration	III	Q	1.0	---	1 0	---	1.0	---	1 0	---	Minute
C1-C3	Stress	IV	S	4.75	5.05	4.75	5 05	4.75	5.05	4.75	5.05	Vrms
D1-D3	Gain check	V	X	0.98	1.02	0 98	1.02	0.97	1.03	0.97	1.03	V
E1-E3	Acquisition time	VI	X	0.0	1.0	0.0	1 2	0.0	1.2	0.0	1.4	µs
F1-F3	Output drift check	VII	X	0.00	3.15	0 00	3.15	0.00	3.47	0.00	3.47	V
H1-H6	Power supply ripple rejection	VIII	Q	0	400	0	400	0	500	0	500	µV
J1,J2	Power supply current	IX	X	5.0	33	5 0	33	5.0	35	5.0	35	mA
K1-K3	Holding capacitor value check	X	S	000963	.001039	.000963	.001039	.000955	.001045	.000955	.001045	µF
L1-L3	Input capacitance	XI	Q	0 0	20	0 0	20	0.0	22	0 0	22	pF
M1-M3	Gate input capacitance	XI	Q	0.0	14	0.0	14	0.0	14	0.0	14	pF
N1-N3	Output crosstalk	XII	Q	0.0	10.0	0.0	10 0	0.0	10.0	0.0	10.0	mVrms
P1,P2	Power supply transient rise time	XIII	S	0 40	0.52	0 40	0.52	0.38	0.54	0.38	0.54	Vpk
Q1-Q3	Input offset voltage	XIV	S	-6.0	+6.0	-6.0	+6.0	-8 0	+8.0	-8.0	+8.0	mV
R1-R3	Aperture time	XV	S	0	100	0	100	0	100	0	100	ns
S1-S3	Second harmonic amplitude	XVI	S	-60	-25	-60	-25	-60	-25	-60	-25	dBV

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through S in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through S except N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through S shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through S shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/191-1 supersedes part number M28787/191. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage and current	+15 Vdc ±10 percent, 35 mA maximum -15 Vdc ±10 percent, 35 mA maximum
Power dissipation	1.16 W maximum
Maximum input signal	5 Vrms
Minimum load resistance	10 kΩ

6.5 Module function. This module contains three sample and hold circuits. Sampling is performed by an analog switch feeding a storage capacitor. The output is buffered by a unity gain operational amplifier to ensure accuracy and provide low-output impedance.

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* 6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.
b Soldering iron tips, metal parts of fixtures and tools, test equipment, and handling facilities should be grounded.
c. Signals should not be applied to the inputs while the device power supply is off.
d. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the protection input diodes.
e All unused input leads should be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate, for the logic circuit involved.

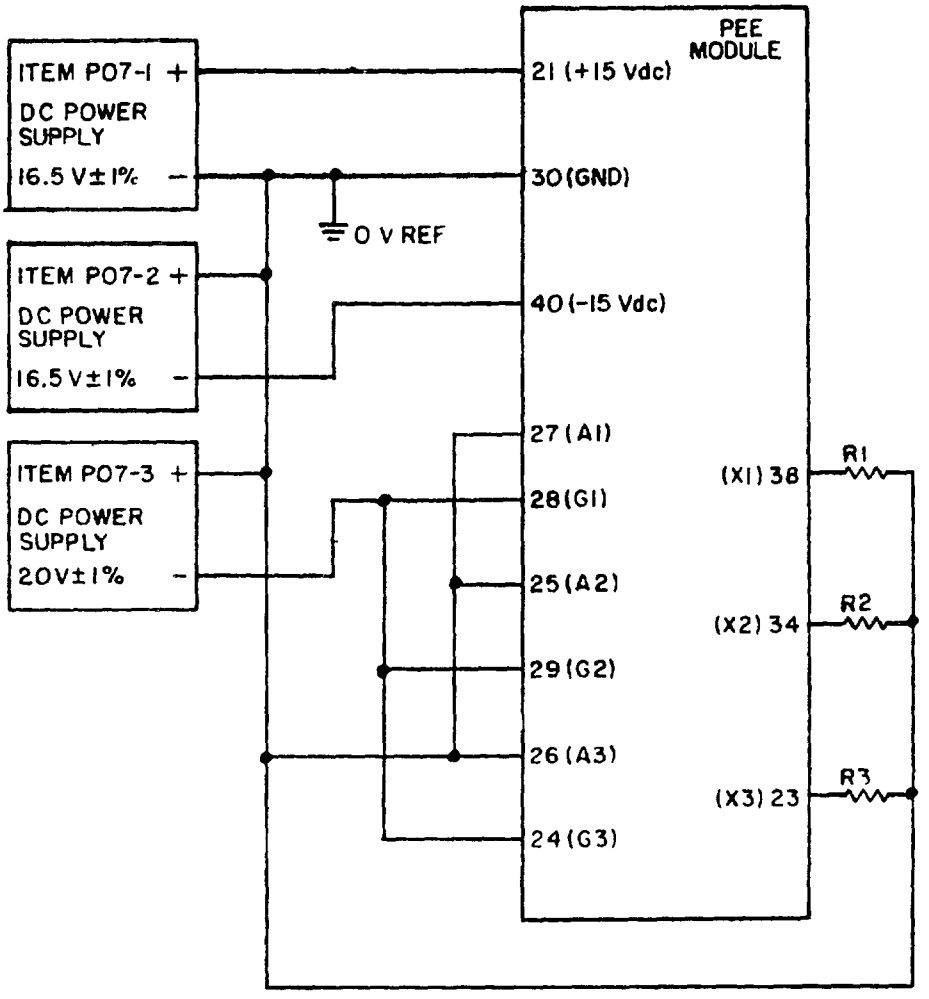
6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES
1. R1-R3 = 10 kΩ ±1%, 1/8 W
2 To prevent module damage turn on items PO7-1 and PO7-2 before connecting to pins 25, 26, and 27. Disconnect pins 25, 26, and 27 first.

* FIGURE 2. Operating temperature, component temperature and life test circuit

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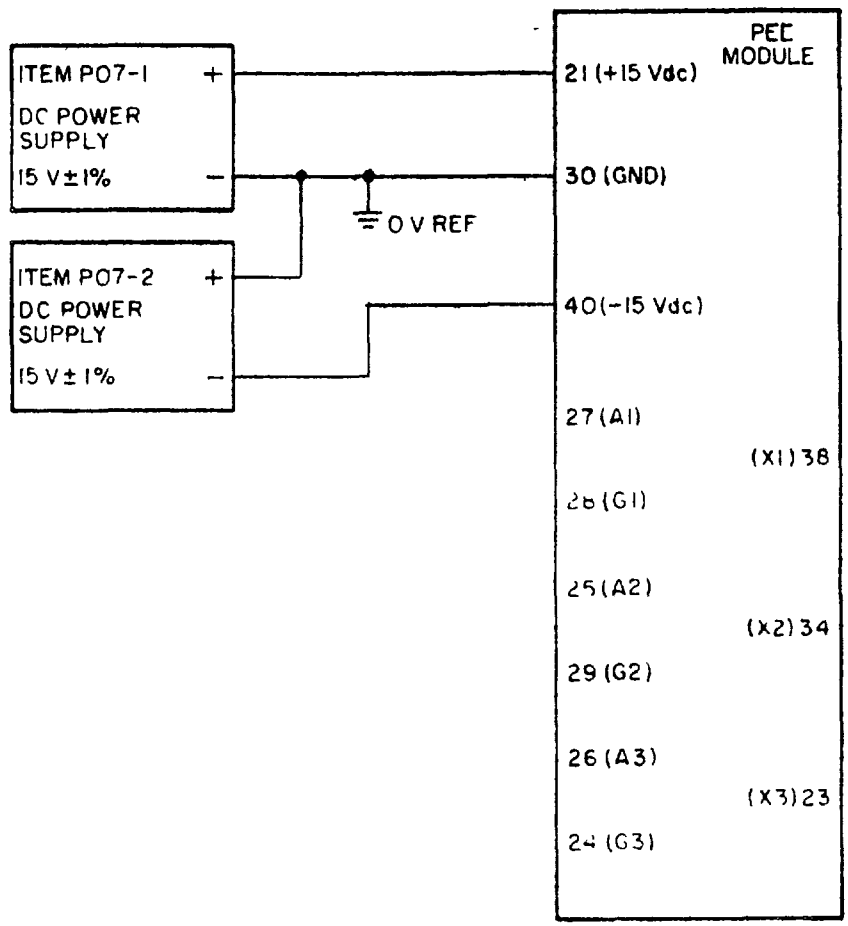


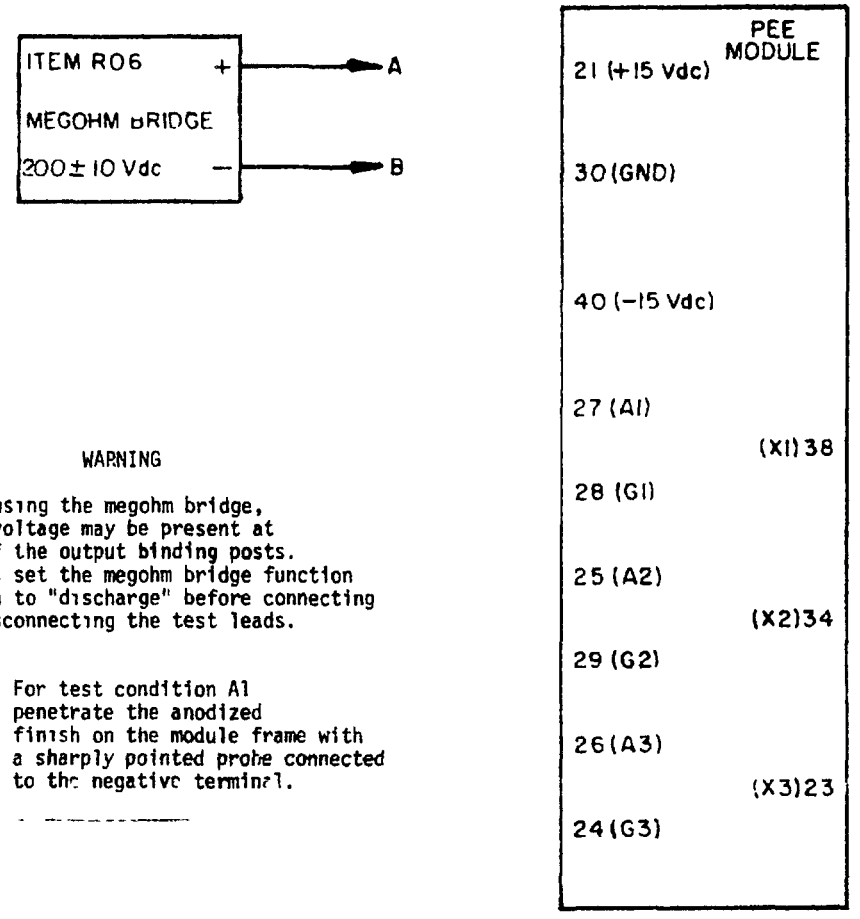
FIGURE 3. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	All pins (including pins 1-20 if supplied)	Frame	10	∞	MΩ	4
A2	Isolation	21,23,24,25, 26,27,28,29, 30,34,38,40	All other pins (including pins 1-20 if supplied)	10	∞	MΩ	4

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

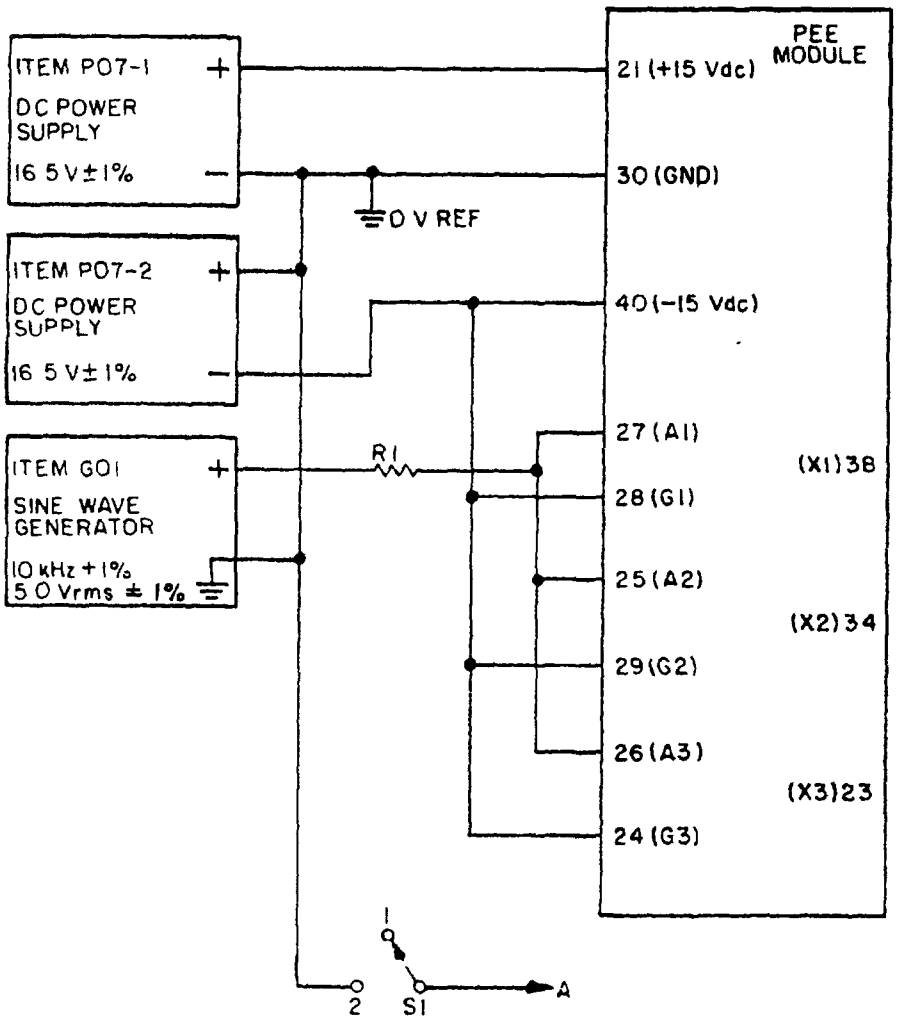
FIGURE 4 Isolation test circuit for table II.

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TABLE III. Output short circuit duration test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	X1 Output short circuit duration	38	1.0	---	Minute	5
B2	X2 Output short circuit duration	34	1.0	---	Minute	5
B3	X3 Output short circuit duration	23	1.0	---	Minute	5

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NOTES:
1. R1 = 2 kΩ ±1%, 1/8 W.
2. S1 = SPST switch.
3. To prevent module damage turn on items P07-1 and P07-2 before connecting item G01 to pins 25, 26 and 27. Disconnect item G01 first.
4. Place switch S1 in position 1 prior to test conditions B1, B2, and E3. Close switch S1 for a minimum of 1 minute for each test condition
5. Satisfactory performance of remaining tests in table I is proof of no damage.

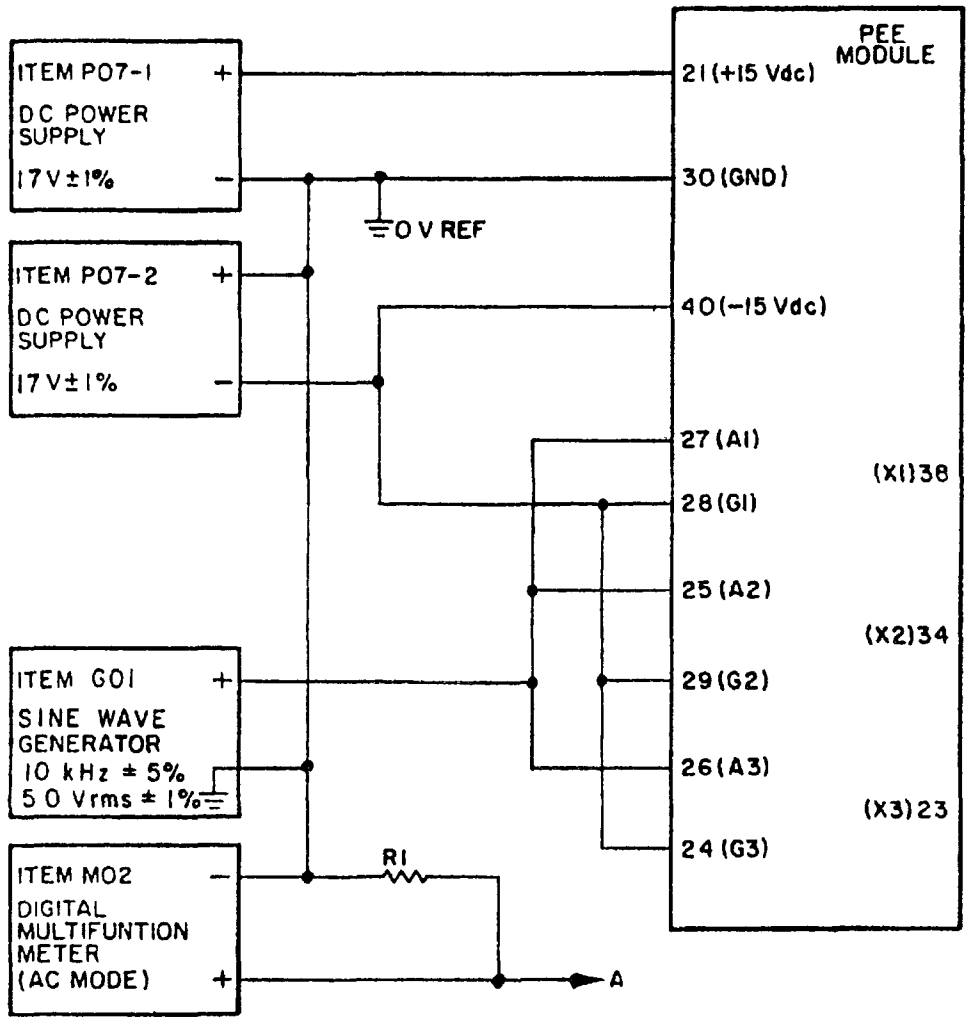
* FIGURE 5 Output short circuit duration test circuit for table III

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TABLE IV. Stress test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
C1	X1	Stress	38	4.75	5.05	Vrms	6
C2	X2	Stress	34	4.75	5.05	Vrms	6
C3	X3	Stress	23	4.75	5.05	Vrms	6

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NOTES
1. R1 = 10 kΩ ±1%, 1/8 W.
2 To prevent module damage turn on items P07-1 and P07-2 before connecting item G01 to pins 25, 26, and 27 Disconnect item G01 first.
3 If these tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

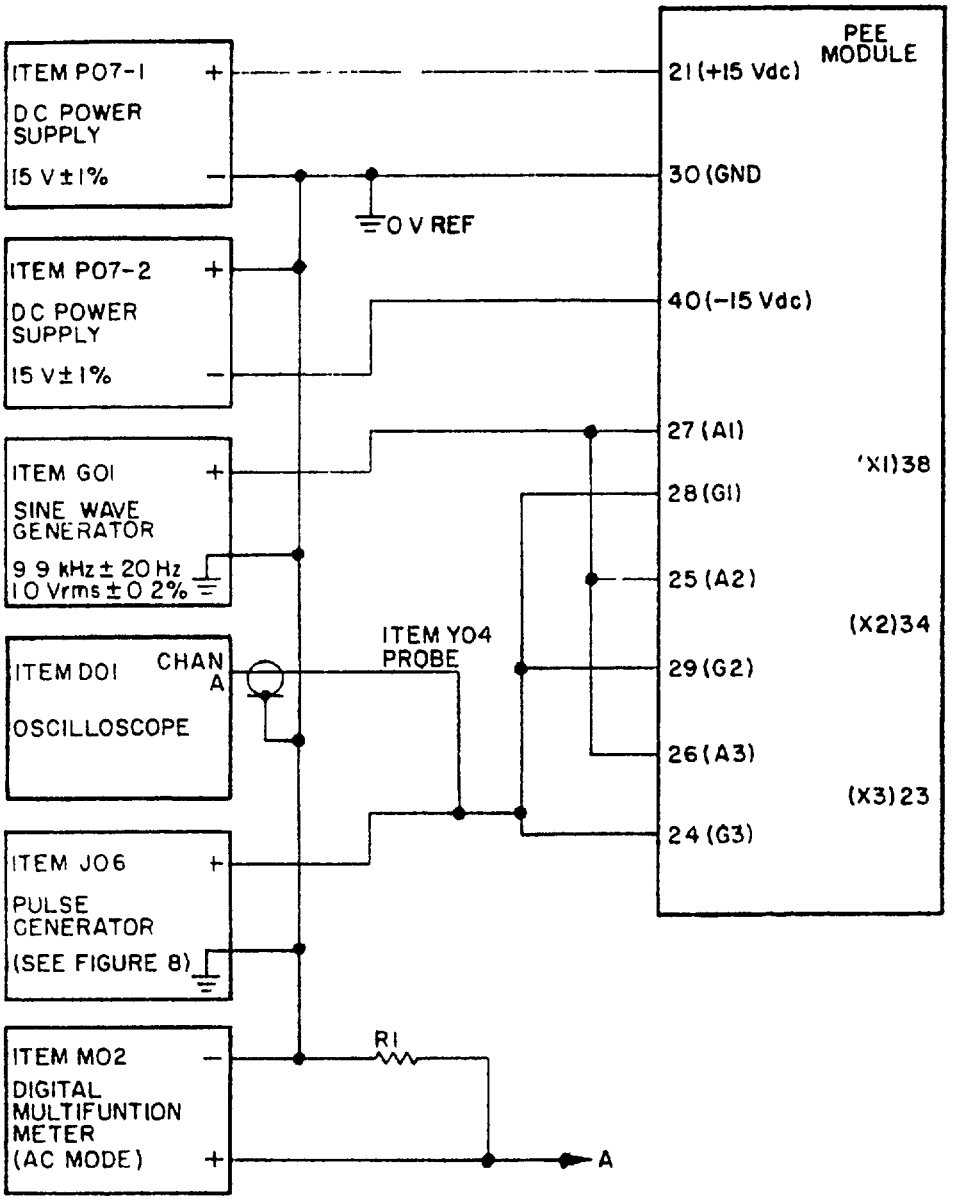
* FIGURE 6 Stress test circuit for table IV

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TABLE V. Gain check test.

Test condition	Parameter		Connect	Initial limits 25° +5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
D1	X1	Gain check	38	0.98	1.02	V	7, 8
D2	X2	Gain check	34	0.98	1.02	V	7, 8
D3	X3	Gain check	23	0.98	1.02	V	7, 8

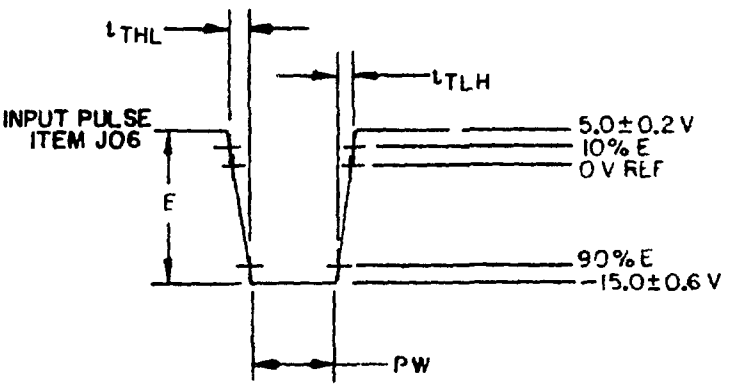
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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W
2. Output impedance of item G01 must be ≤ 50Ω
3. To prevent module damage turn on items P07-1 and P07-2 before connecting item G01 to pins 25, 26 and 27. Disconnect item G01 first.
4. The frequency of the signal at point A will be the difference between the frequencies of item J06 and item G01 (approximately 100 Hz).

* FIGURE 7. Gain check test circuit for table V.

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NOTE: Input pulse parameters:
PW = 200 ±50 ns.
$t_{TLH} = t_{THL} \leq 50$ ns.
PRR = 10kHz ±20 Hz

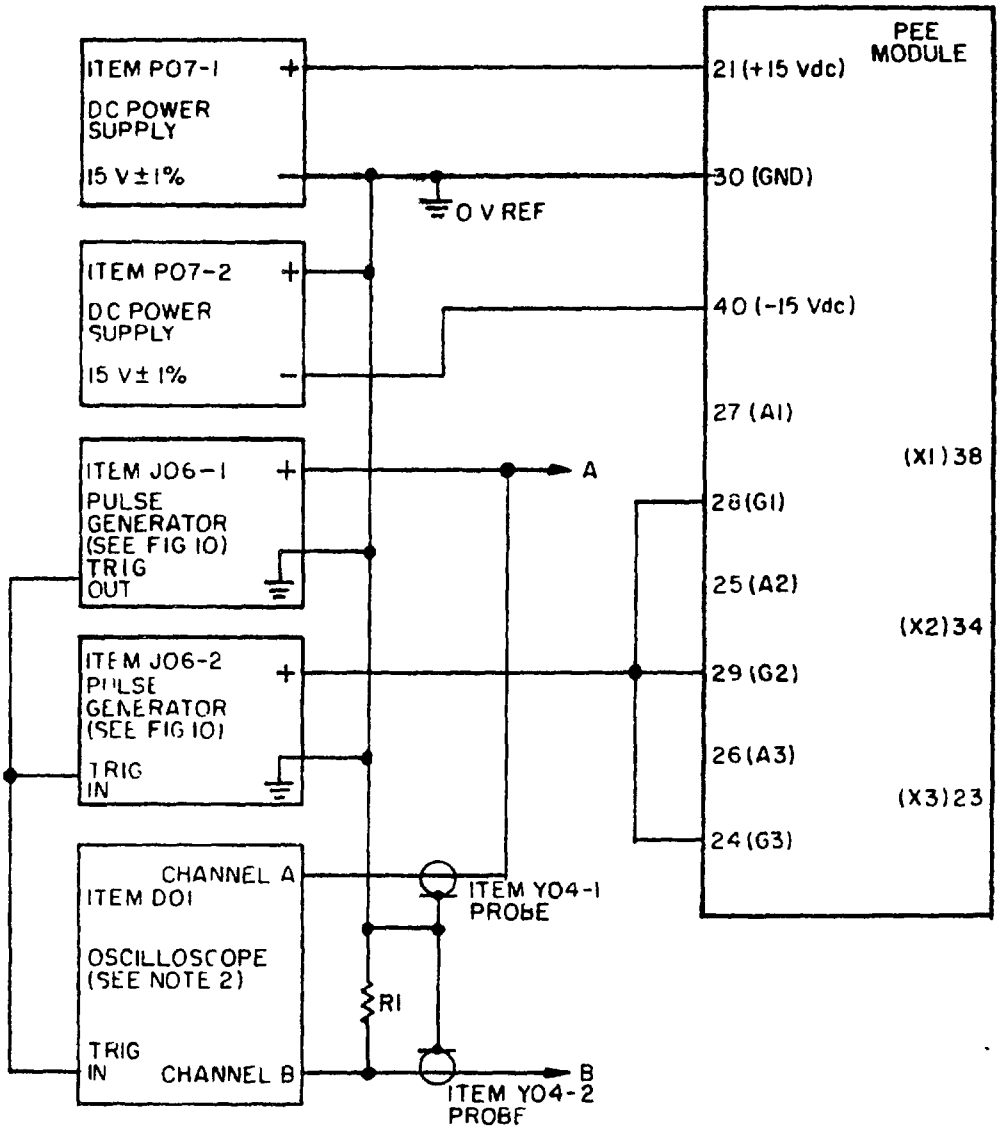
FIGURE 8. Gain check test circuit waveform for table V.

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TABLE VI. Acquisition time test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
E1	A1	Acquisition time	27	38	0.0	1.0	µs	9,10
E2	A2	Acquisition time	25	34	0.0	1.0	µs	9,10
E3	A3	Acquisition time	26	23	0.0	1.0	µs	9,10

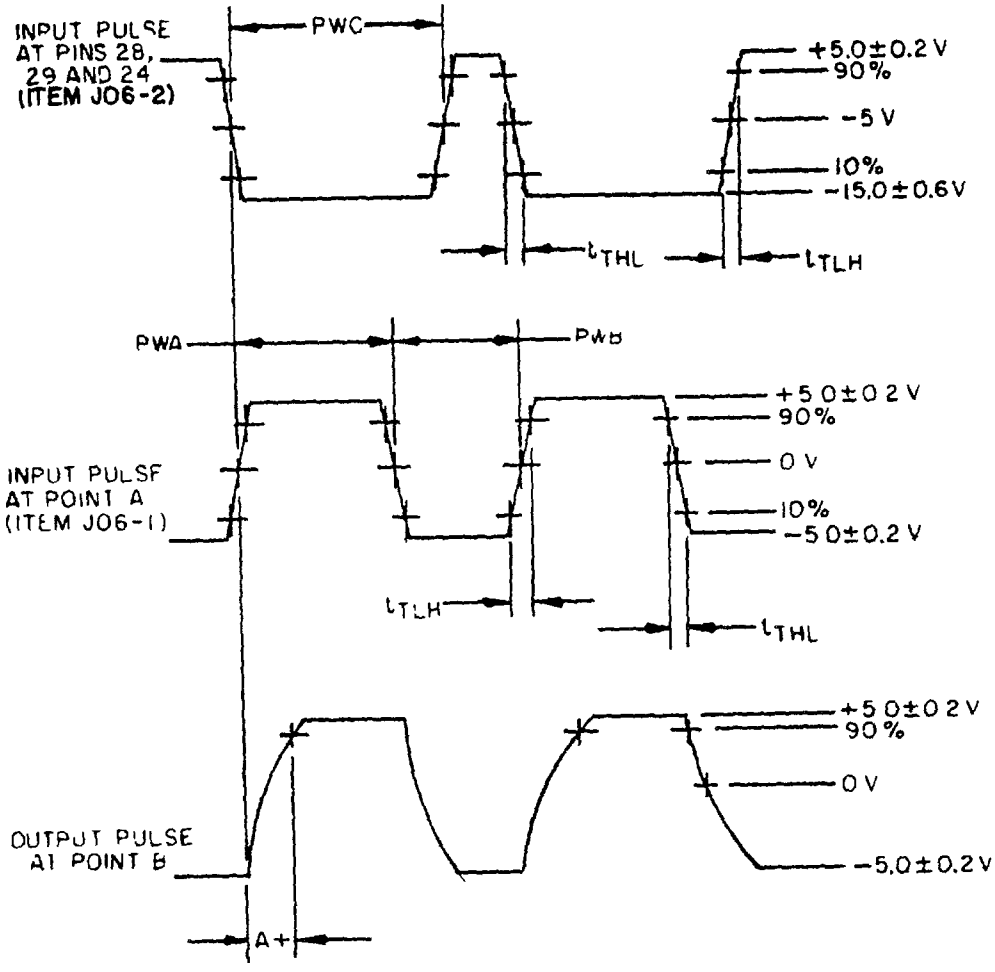
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NOTES
1. R1 = 10 kΩ ±1%, 1/8 W.
2. Trigger item D01 on the leading edge of the waveform at point A (see figure 10).
3 To prevent module damage turn on items P07-1 and P07-2 before connecting item J06-1 to pin 25, 26, or 27. Disconnect item J06-1 first

* FIGURE 9. Acquisition time test circuit for table VI.

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NOTES
1. Input pulse parameters (item J06-1):
   PWA = PWB = 50 ±5 µs.
   PRR = 10 kHz ±5%.
   $t_{TLH} = t_{THL} \leq 50$ ns.
2. Input pulse parameters (item J06-2):
   PWC = 75 ±5 µs
   PRR = Triggered by item J06-1.
   $t_{TLH} = t_{THL} \leq 50$ ns.
3 A+ = Acquisition time

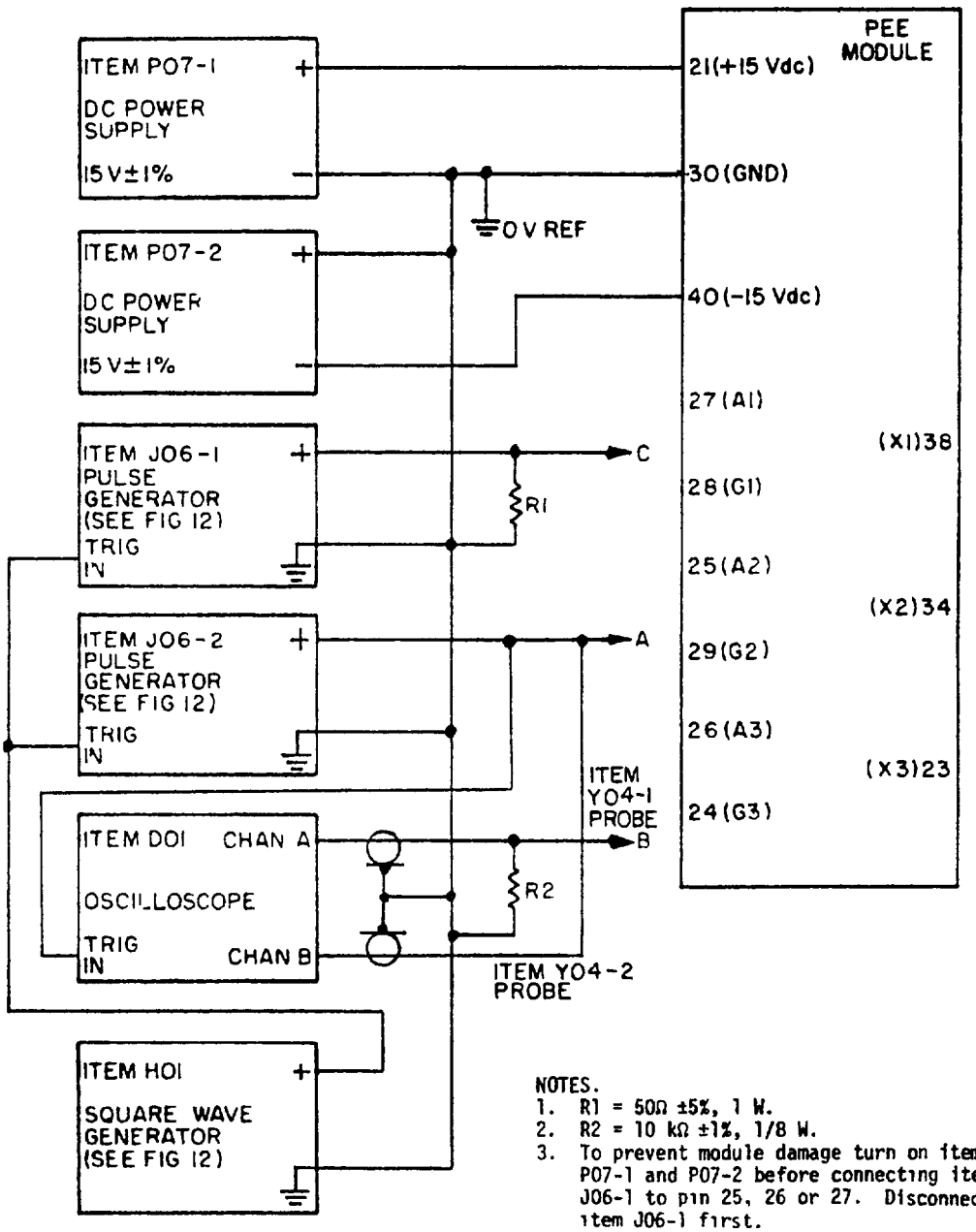
* FIGURE 10 Acquisition time test circuit waveforms for table VI

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TABLE VII. Output drift check test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	X1 Output drift check	28	38	27	0.00	3.15	V	11,12
F2	X2 Output drift check	29	34	25	0.00	3.15	V	11,12
F3	X3 Output drift check	24	23	26	0.00	3.15	V	11,12

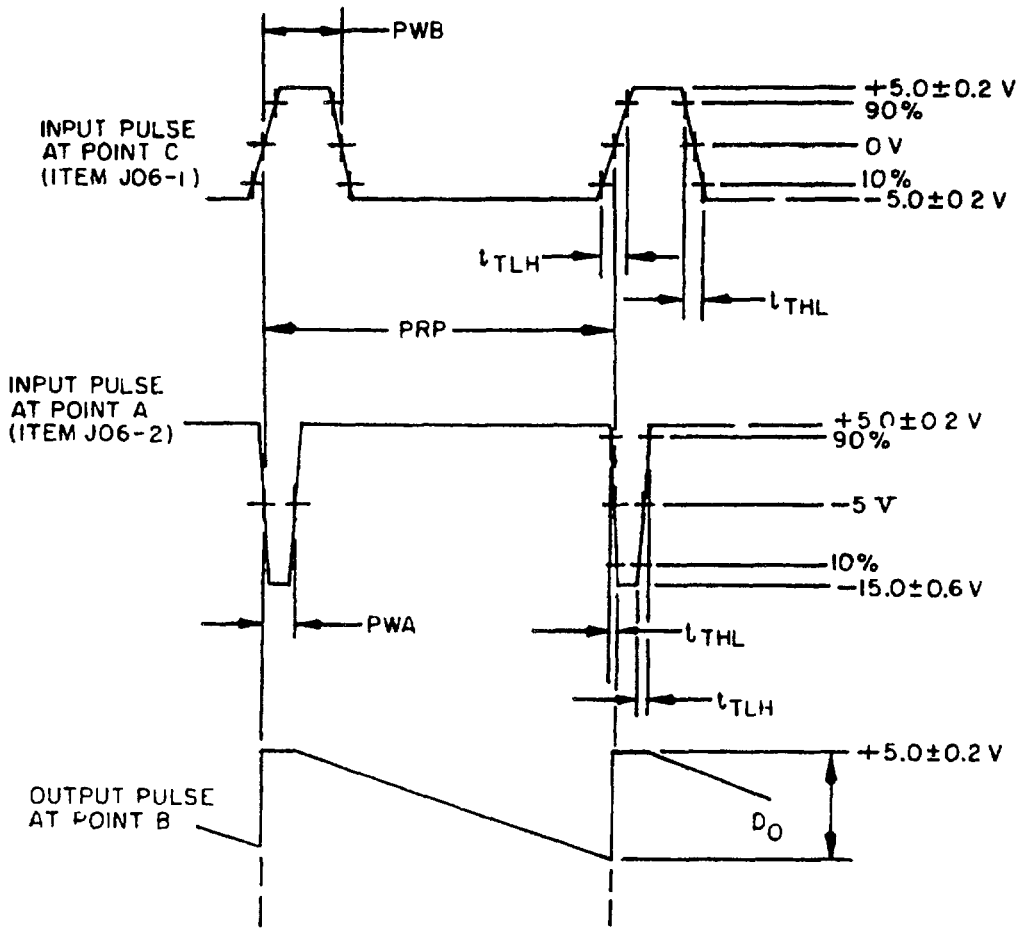
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NOTES.
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 10 kΩ ±1%, 1/8 W.
3. To prevent module damage turn on items PO7-1 and PO7-2 before connecting item JO6-1 to pin 25, 26 or 27. Disconnect item JO6-1 first.

★ FIGURE 11 Output drift check test circuit for table VII

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NOTES:
1. Input pulse parameters (item J06-1).
   PWB = 3.0 ±0.3 ms.
   PRP = 1.0 ±0.1 s (triggered by item H01).
   $t_{TLH} = t_{THL} \leq 50$ ns.
2. Input pulse parameters (item J06-2).
   PWA = 1.0 ±0.1 ms.
   PRP = 1.0 ±0.1 s (triggered by item H01).
   $t_{TLH} = t_{THL} \leq 50$ ns
3. Item H01 pulse parameters.
   Frequency = 1 Hz ±10%.
   Amplitude = 5 V_{p-p} +6%.
4. D_0 = Output drift voltage

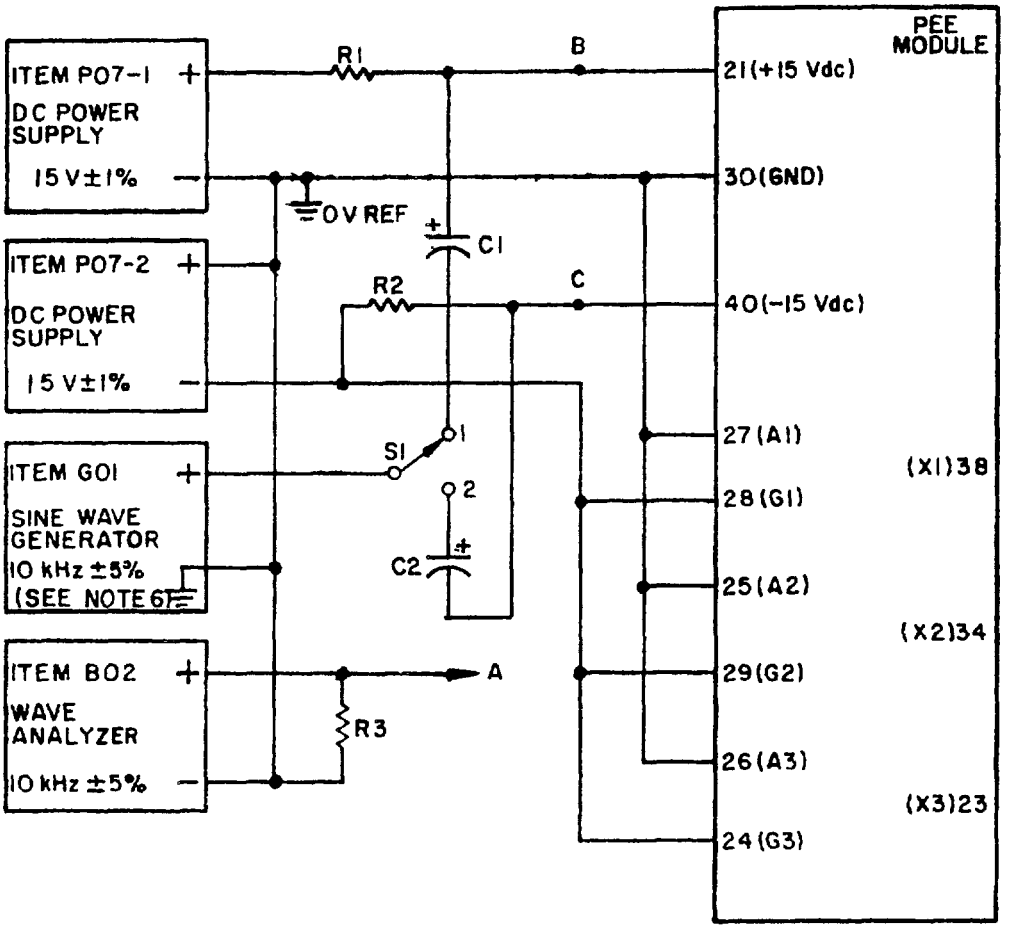
* FIGURE 12. Output drift check test circuit waveforms for table VII.

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TABLE VIII Power supply ripple rejection test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		\ to pin	Min	Max		
H1	Power supply ripple rejection	38	0	400	µV	13
H2		38				
H3		34				
H4		34				
H5		23				
H6	Power supply ripple rejection	23	0	400	µV	13

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NOTES:
1. R1 = R2 = 47Ω ±5%, 1/4 W.
2. R3 = 10 kΩ ±1%, 1/8 W.
3. C1 = C2 = 20 µF ±20%, 20 Vdc.
4. S1 = SPDT switch.
5. Set switch S1 to position 1 for test conditions H1, H3, and H5 and to position 2 for test conditions H2, H4 and H6
6. Adjust item G01 for 150 mVrms ±1% at point B or point C as applicable.
7. To prevent module damage, turn on items P07-1 and P07-2 before connecting to pins 25, 26, and 27 Disconnect pins 25, 26, and 27 first.

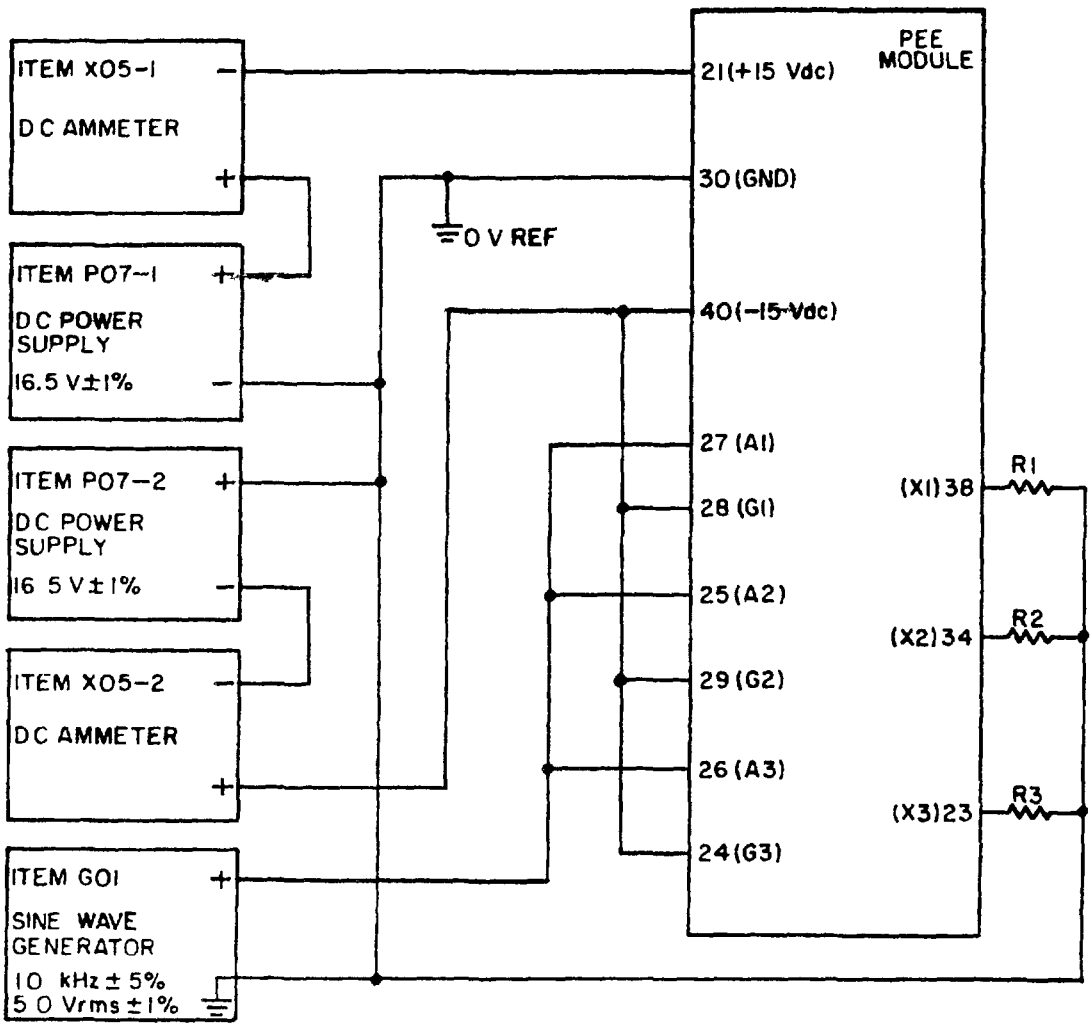
★ FIGURE 13 Power supply ripple rejection test circuit for table VIII.

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TABLE IX. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
J1	$+V_{CC}$ Power supply current	5.0	33	mA	14
J2	$-V_{CC}$ Power supply current	5.0	33	mA	14

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NOTES
1. R1 - R3 = 10 kΩ ±1%, 1/8 W.
2. For test condition J1 read item XO5-1 and for test condition J2 and read item XO5-2.
3. To prevent module damage, turn on items PO7-1 and PO7-2 before connecting item GO1 to pins 25, 26, and 27. Disconnect item GO1 first.

* FIGURE 14. Power supply current test circuit for table IX

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* TABLE X Holding capacitor value check test. 1/

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K1	A1	Holding capacitor value check	27	38	10.15	10.94	kHz	15
K2	A2	Holding capacitor value check	25	34	10.15	10.94	kHz	15
K3	A3	Holding capacitor value check	26	23	10.15	10.94	kHz	15

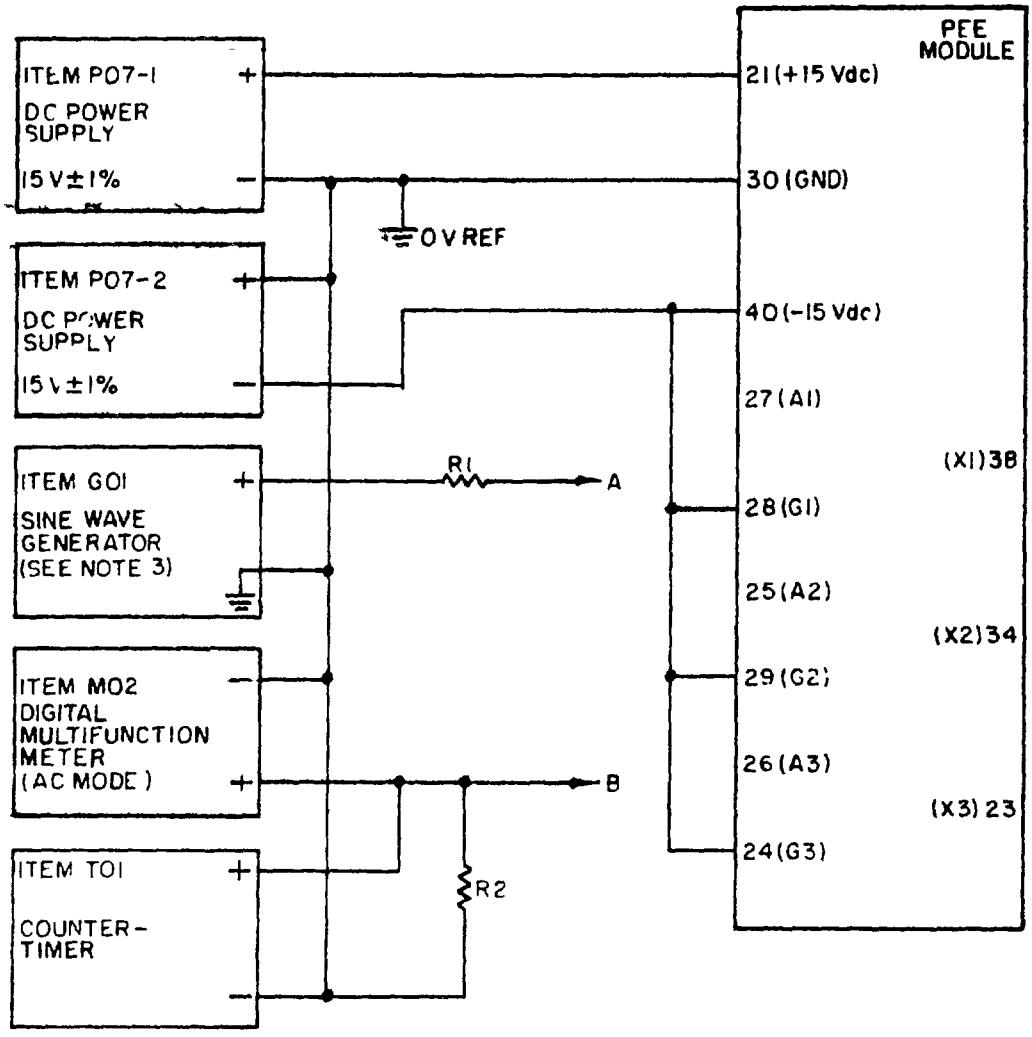
1/ This test verifies the holding capacitor value to be 0.001 µF ±3.8 percent at 25°C in accordance with the following relationship:

$$C = \left[\left(\frac{1}{2\pi\,(R1 + FET\ R_{ON})\ (\text{test condition K reading})}\right) - (C_{FET} + C_{STRAY})\right] \text{Farads}$$

or $$C = \left[\left(\frac{1}{2\pi\,(14875\Omega)\ (\text{test condition K reading})}\right) - (15\ \text{pF})\right] \text{Farads}$$

This assumes R_{ON} to be 175Ω and other variables to be nominal.

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NOTES:
1. R1 = 14.7 kΩ ±0.1%, 1/8 W.
2. R2 = 10 kΩ ±1%, 1/8 W.
3. Each test condition shall be performed as follows:

   a. Make required connections.
   b. Set frequency of item G01 to 200 Hz (as read on item T01).
   c. Vary the amplitude of item G01 until item M02 (at point B) reads 5.00 ±0.01 Vrms. Record the voltage at item G01 output.
   d. Increase the frequency of item G01 until item M02 reads 3.535 Vrms (3 dB down). Readjust the voltage at item G01 output if necessary to equal the voltage recorded in step c. The frequency read on item T01 shall meet the limits of table X.

4. To prevent module damage, turn on items P07-1 and P07-2 before connecting item G01 to pin 25, 26, or 27. Disconnect item G01 first.

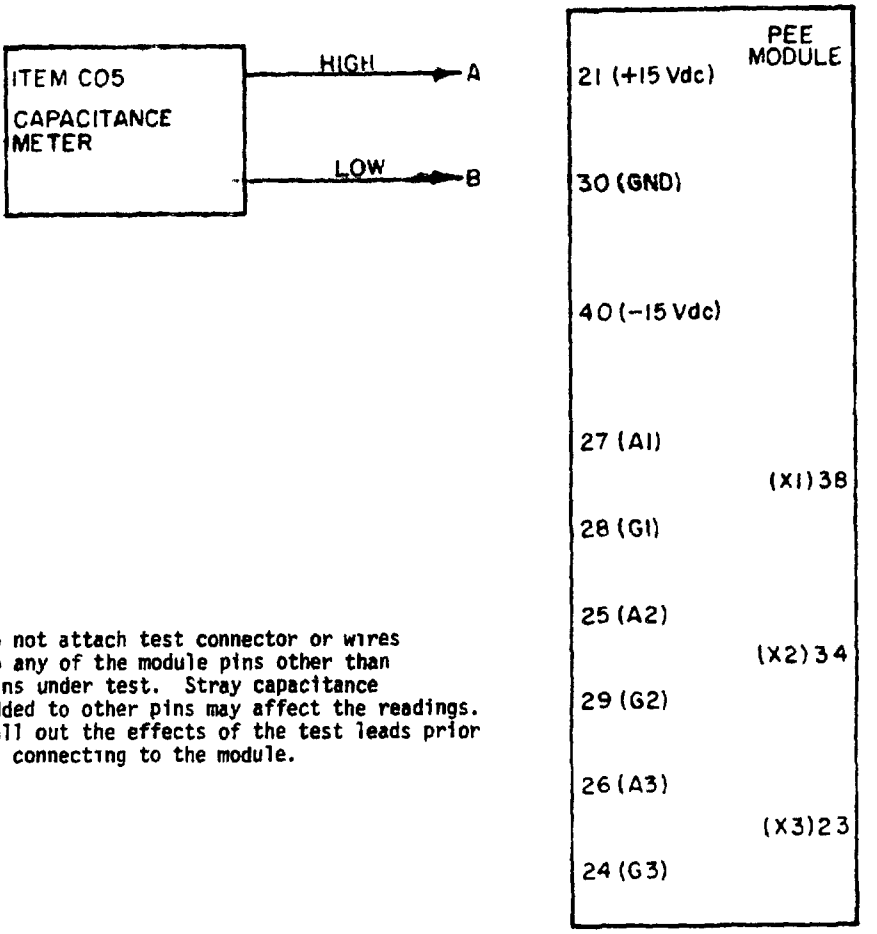
* FIGURE 15. Holding capacitor value check test circuit for table X.

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TABLE XI. Input capacitance and gate input capacitance test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	A1	Input capacitance	27	30	0.0	20	pF	16
L2	A2	Input capacitance	25	↓	↓	20	↓	↓
L3	A3	Input capacitance	26	↓	↓	20	↓	↓
M1	G1	Gate input capacitance	28	↓	↓	14	↓	↓
M2	G2	Gate input capacitance	29	↓	↓	14	↓	↓
M3	G3	Gate input capacitance	24	30	0.0	14	pF	16

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pins under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting to the module.

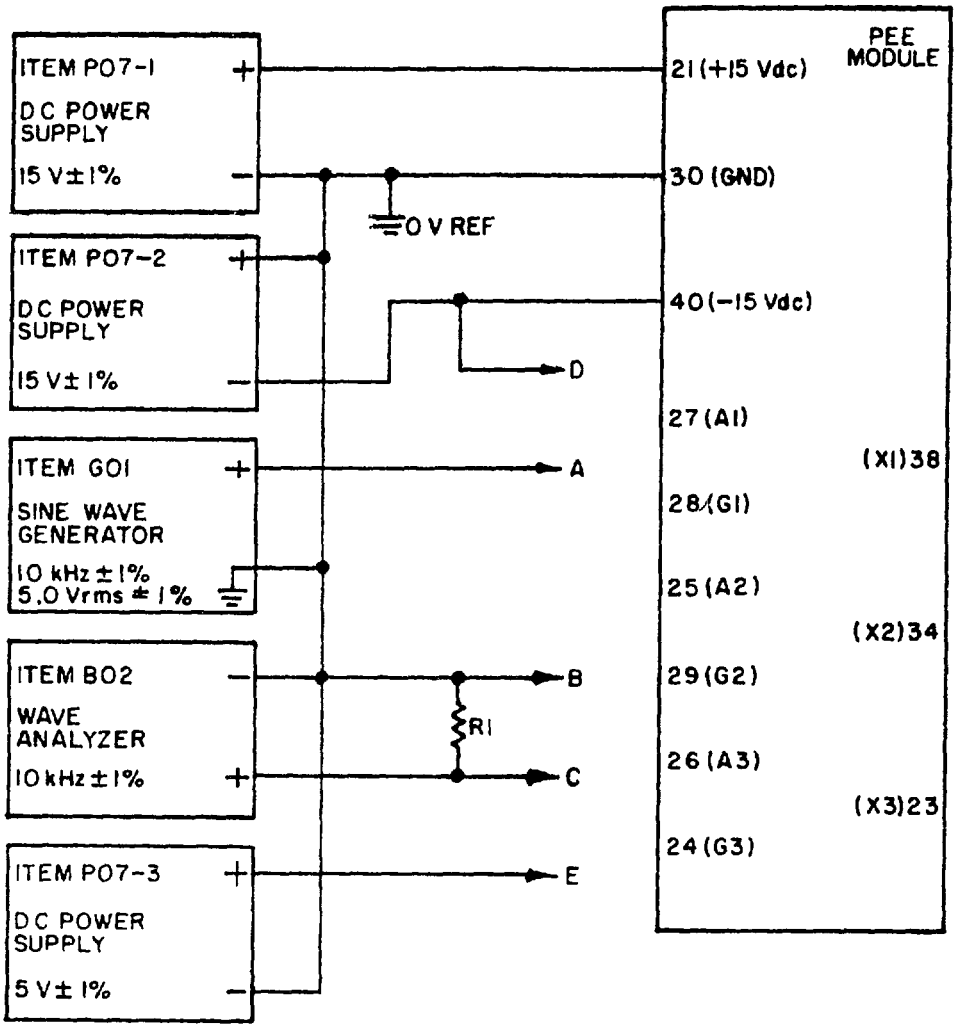
FIGURE 16. Input capacitance and gate input capacitance test circuit for table XI

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TABLE XII. Output crosstalk test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N1	Output crosstalk	27,25	26	23	28,29	24	0.0	10.0	mVrms	17
N2	Output crosstalk	25,26	27	38	24,29	28	0.0	10.0	mVrms	17
N3	Output crosstalk	27,26	25	34	24,28	29	0.0	10.0	mVrms	17

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. To prevent module damage turn on items P07-1 and P07-2 before connecting item G01 to pin 25, 26, or 27. Disconnect item G01 first.

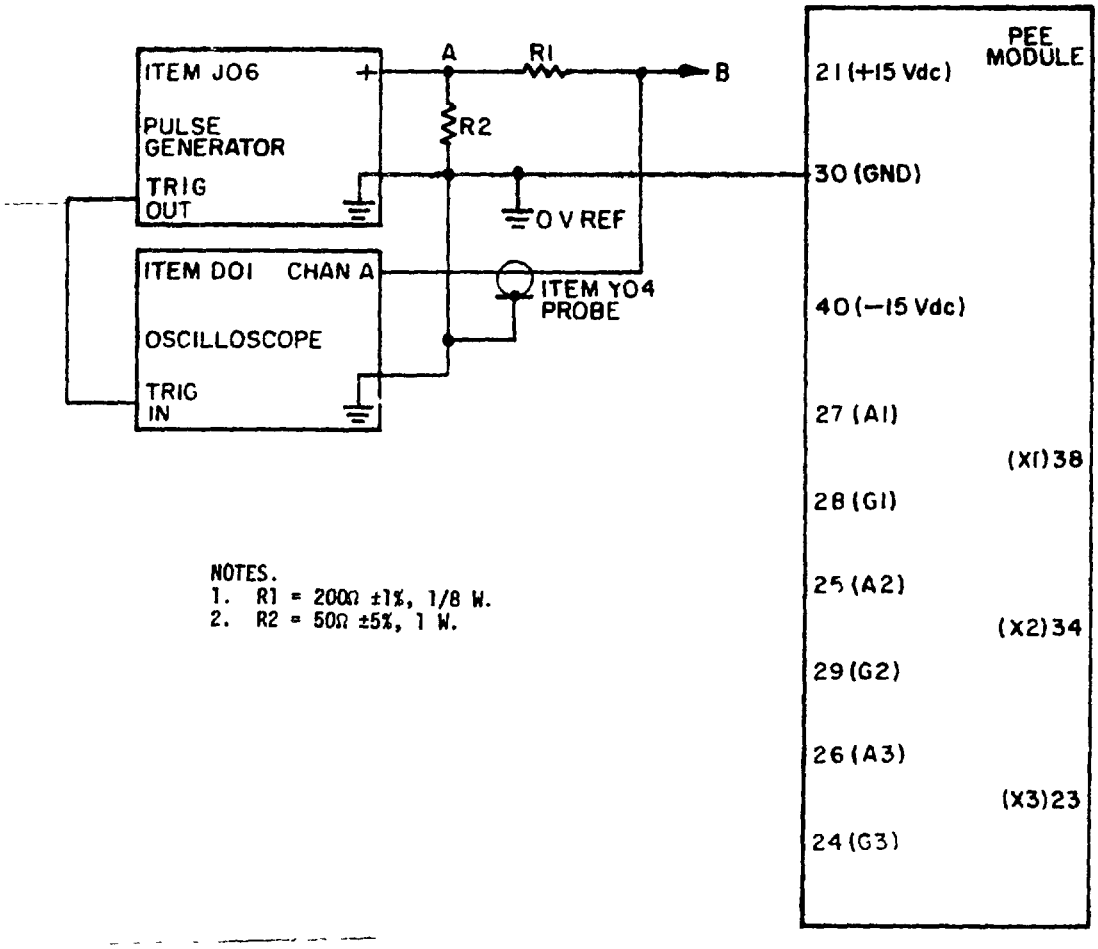
★ FIGURE 17. Output crosstalk test circuit for table XII.

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TABLE XIII. Power supply transient rise time.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		B to pin	Min	Max		
P1	$+V_{CC}$ Power supply transient rise time	21	0.40	0.52	Vpk	18,19
P2	$-V_{CC}$ Power supply transient rise time	40	0.40	0.52	Vpk	18,19

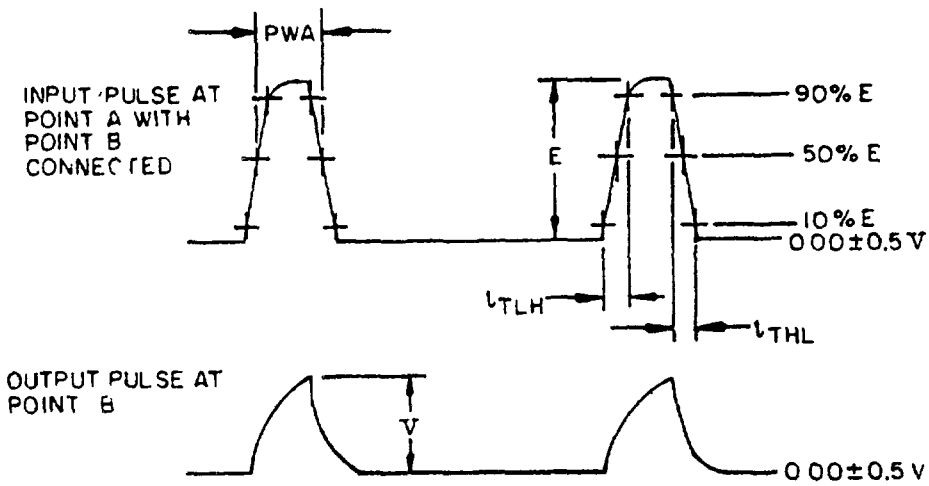
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NOTES.
1. R1 = 200Ω ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.

FIGURE 18 Power supply transient rise time test circuit for table XIII.

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NOTES
1. Input pulse parameters:
   E = 0.60 ±0.03 V.
   PWA = 50 ±2.5 µs.
   PRP = 600 µs ±10%.
   $t_{TLH} = t_{THL}$ = 50 ns maximum.
2. Measure amplitude V at Point B and compare to table XIII limits

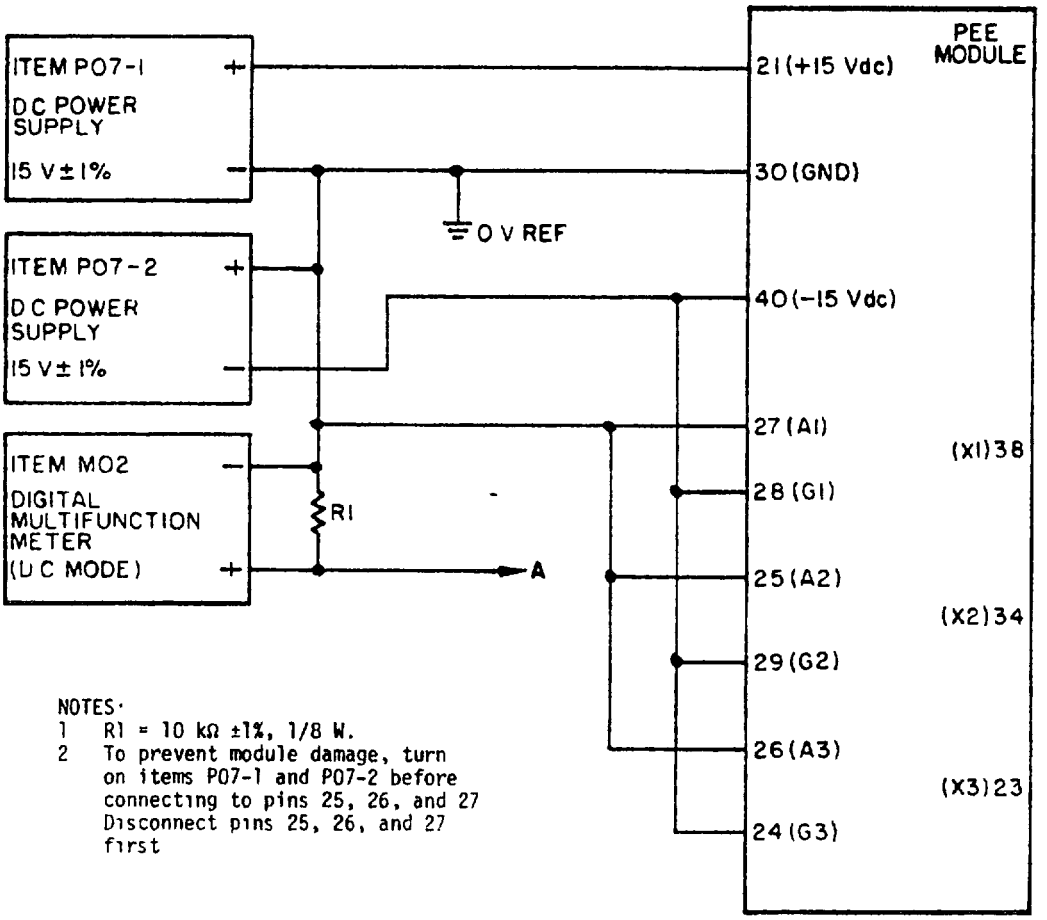
FIGURE 19 Power supply transient rise time test circuit waveforms for table XIII.

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TABLE XIV. Input offset voltage test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	Input offset voltage	38	-6.0	+6.0	mV	20
Q2	Input offset voltage	34	-6.0	+6.0	mV	20
Q3	Input offset voltage	23	-6.0	+6.0	mV	20

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. To prevent module damage, turn on items P07-1 and P07-2 before connecting to pins 25, 26, and 27. Disconnect pins 25, 26, and 27 first.

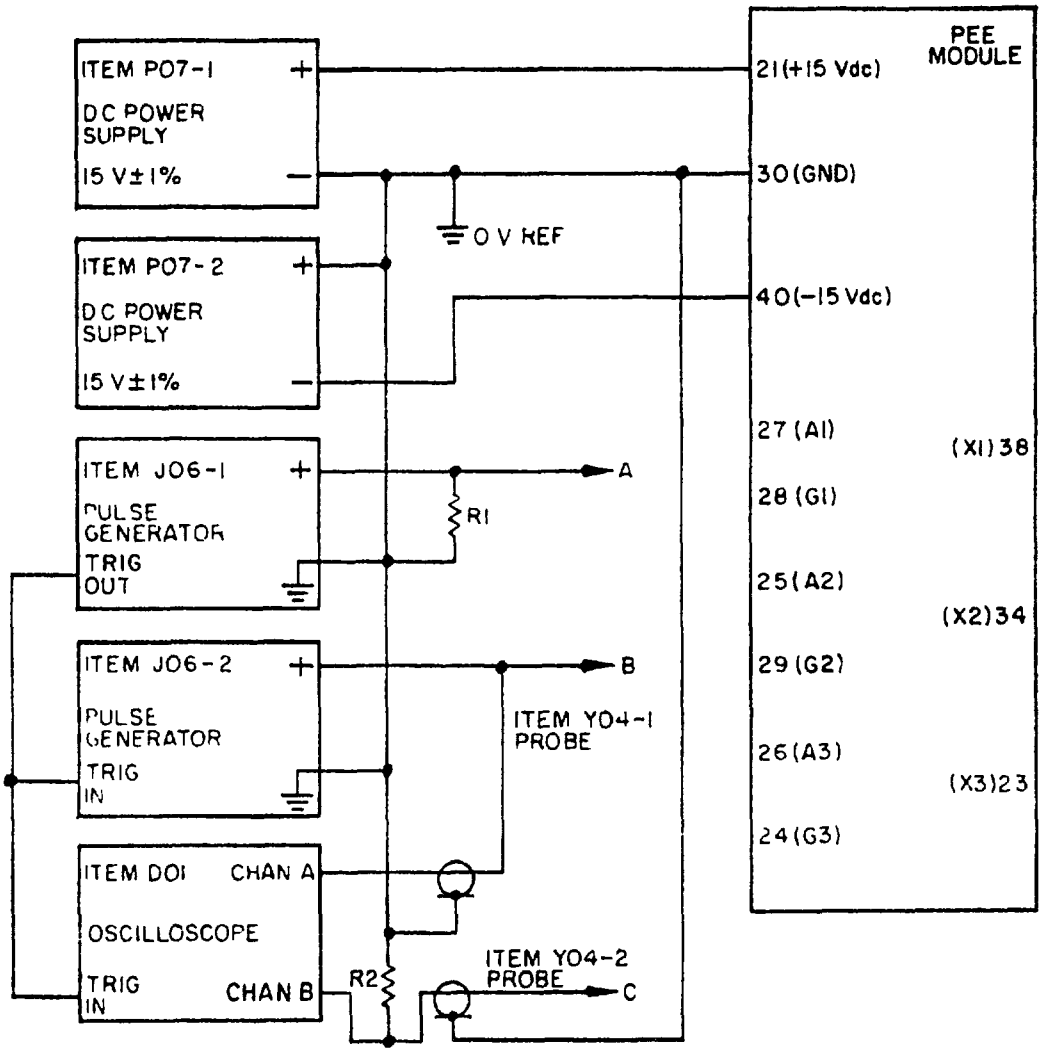
* FIGURE 20. Input offset voltage test circuit for table XIV.

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TABLE XV. Aperture time test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R1	A1	Aperture time	27	28	38	0	100	ns	21,22
R2	A2	Aperture time	25	29	34	0	100	ns	21,22
R3	A3	Aperture time	26	24	23	0	100	ns	21,22

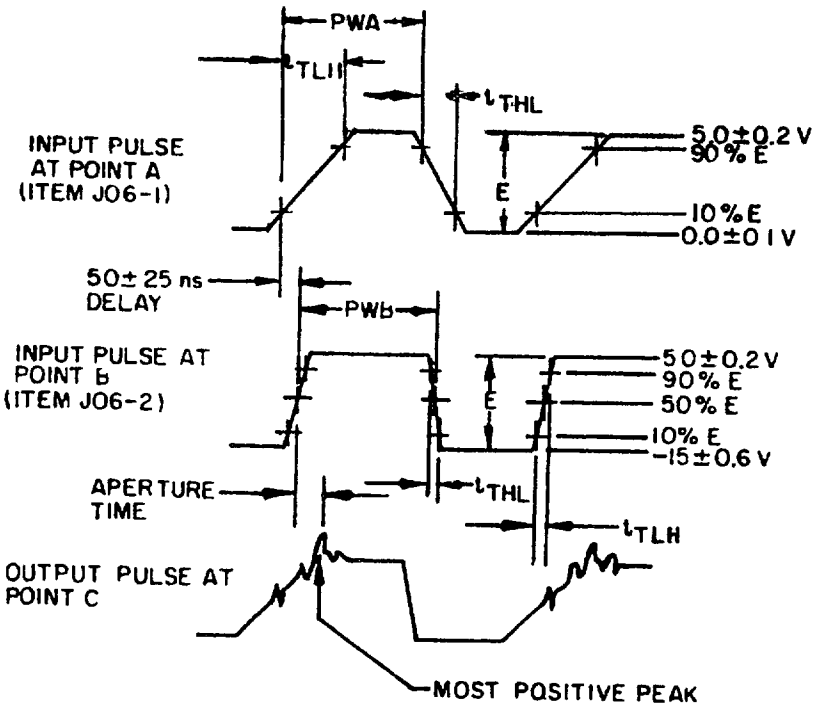
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NOTES
1. R1 = 50Ω ±5%, 1 W
2. R2 = 10 kΩ ±1%, 1/8 W
3 Connect the ground of item Y04-2 probe as close as possible to pin 30
4 To prevent module damage, turn on items P07-1 and P07-2 before connecting item J06-1 to pins 25, 26, and 27. Disconnect item J06-1 first

* FIGURE 21 Aperture time test circuit for table XV

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NOTES.
1. Input pulse parameters at point A:
   PWA = 10.0 ±0.5 µs.
   PRF = 50 kHz ±10%.
   $t_{TLH} = t_{THL} = 200 \pm 20$ ns.
2. Input pulse parameters at point B:
   PWB = 11.0 ±0.5 µs.
   PRR = Triggered by item J06-1.
   $t_{TLH} = t_{THL} \leq 20 \pm 5$ ns.

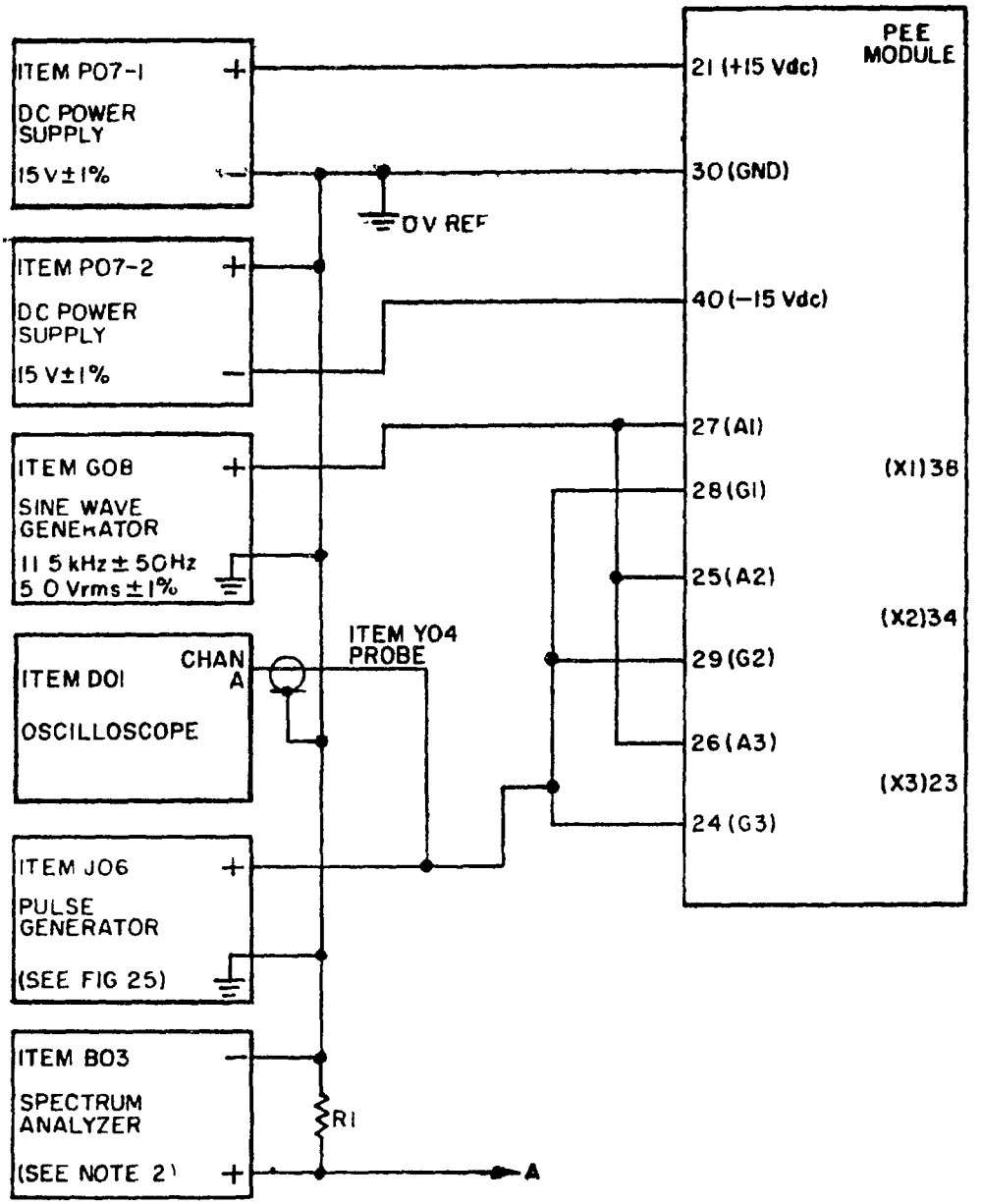
* FIGURE 22. Aperture time test circuit waveforms for table XV.

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TABLE XVI. Second harmonic amplitude test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
S1	Second harmonic amplitude	38	-60	-25	dBV	23,24
S2	Second harmonic amplitude	34	-60	-25	dBV	23,24
S3	Second harmonic amplitude	23	-60	-25	dBV	23,24

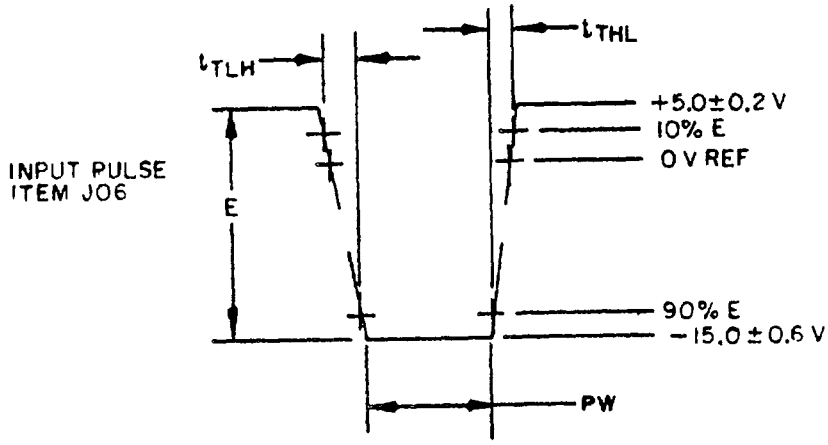
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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W
2. Set item B03 to
   Amplitude mode - Log dB/Div
   Bandwidth - 10 Hz
   Sweep mode - 0.1 kHz/Div
   Sweep from 100 Hz to 1.1 kHz and record
   Peak amplitude at 800 ±200 Hz (at double output frequency at 400 ±100 Hz fundamental).
3. To prevent module damage turn on items P07-1 and P07-2 before connecting item G08 to pins 25, 26, and 27. Disconnect item G08 first.

★ FIGURE 23 Second harmonic amplitude test circuit for table XVI.

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NOTE: Input pulse parameters:
PW = 300 ±50 ns.
$t_{TLH} = t_{THL} \leq 50$ ns.
PRR = 11.1 kHz ±50 Hz.

FIGURE 24. Second harmonic amplitude test circuit waveform for table XVI.

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